

(hand out) Exponents, Scientific Notation

Exponents

What is the difference between

$$-6^2 \quad \text{versus} \quad (-6)^2$$

This is from the HW:

What is the exponent rule(s) that you would use to simplify?

Simplify means no parenthesis, no negative exponents

$$(-3v^8)^2$$

$$(x^3)^{11}$$

$$(6w^7)(10w^8)$$

$$(5x^2)^0$$

$$\frac{a^7}{a^2}$$

$$\left(\frac{y^3}{4}\right)^4$$

Scientific Notation

Which numbers are in correct scientific notation?

Order the numbers from small to big

Write some as expanded numbers (AKA "decimal")

$$3 \times 10^8$$

$$6.022 \times 10^{23}$$

$$0.5 \times 10^{-3}$$

$$98 \times 10^5$$

$$3.14$$

$$10 \times 2^{10}$$

$$6.62607 \times 10^{-34}$$

$$0.001 \times 10^{-20}$$

$$100 \times 10^6$$

When you multiple/divide numbers in scientific notation, create two groups: combine the coefficients together, combine the 10's together.

Check: are the coefficients a single digit....AKA is the number on the interval $[1,10)$?

If the number is less than 1, give the coefficient more powers of 10 (and decrease the exponent on the 10)

If the number is 10 or more, remove powers of 10 from the coefficient (and increase the exponent on the 10)

Scientific Notation HW problems

Paige saw the number shown on her calculator screen.



Which number represents the number Paige saw?

- ☐ -6,000,000
- ☐ 0.000 06
- ☐ -600,000
- ☐ 0.000 006

How many times larger than 2×10^3 is 6×10^{18} ?

- ☐ 3×10^6
 - ☐ 3×10^{15}
 - ☐ 3×10^{21}
 - ☐ 3×10^3
-

Multiply the following and express your answer in appropriate scientific notation:

$$(8.7 \times 10^{-20}) \times (1.7 \times 10^{36})$$

How many centimeters are in 378.25 meters? Answer in scientific notation.

You need to convert a runners time from $1.53 \cdot 10^6$ milliseconds to to seconds for a track coach. Answer in scientific notation.